

Passive Pre-Association Wi-Fi Access-Point Selection

Alistair

Wi-Fi AP Selection Research Project

Abstract— A Wi-Fi client in overlapping coverage must select an access point (AP) before it can directly measure the service that each candidate would provide. Signal strength describes link quality, but not contention or load, and may therefore favor a nearby but congested AP. This project investigates whether a client can make a better decision using only passive observations collected before association, without probe requests or assistance from the network. This report defines the experimental foundation of that investigation: a controlled ns-3 simulator that creates multi-AP deployments, records decoded frames and physical carrier-sense activity, and replays each fixed topology once per candidate AP, changing only the AP the candidate joins. Every replay of a topology therefore shares one identical pre-association observation while yielding its own measured throughput. The simulator separates physical-topology randomness from radio randomness and produces a structured choice-set dataset. The resulting method supports a reproducible comparison between conventional signal-based selection and selection based on richer passive evidence.

I. PROBLEM FORMULATION

Throughput depends on channel contention, competing stations, transmission rates, retries, and the airtime consumed by neighboring basic service sets. Received signal strength reflects none of them directly. Those effects do, however, leave traces a client can hear without transmitting: the frames themselves, and the time the medium spends busy. This work asks whether such traces are enough: *can a client choose the AP that will provide the greatest throughput using only what its radio can passively observe before association?*

Existing load-aware methods [1], [2] make their load estimate after association, from AP-provided reports, or from a network controller. A client making its first choice has none of these: only beacons and some data frames, in which AP distance and offered traffic appear only through their effects.

Let $\mathcal{A} = \{a_1, \dots, a_N\}$ be the APs in one deployment and let O be the candidate station's passive observation before it joins any AP. If y_i is the throughput obtained after joining a_i , the desired decision is

$$a^* = \arg \max_{a_i \in \mathcal{A}} y_i, \quad (1)$$

using a decision function whose inputs are derived only from O . In a live network only one y_i is ever measurable: the station joins a single AP and observes that AP's throughput, while the values of y_j for every AP it did not join are counterfactual. The simulator recovers all N values by replaying the same scenario

once per target AP, changing only the AP the candidate joins. Every choice set therefore carries its true a^* .

The replay is sound only if O is identical across all N runs of a scenario. If any part of the observation varied with the AP eventually joined, a decision function could reach the right answer through a simulation artifact rather than through the network. This is checked rather than assumed. The assumptions that do remain are stated next.

II. EXPERIMENTAL SCOPE AND ASSUMPTIONS

Simulator 1 isolates the relationship among path quality, background load, and passive evidence. It does not attempt to reproduce every feature of a deployed WLAN. Table I states the principal assumptions so that conclusions can be interpreted within the intended scope.

TABLE I
CONTROLLED ASSUMPTIONS OF SIMULATOR 1.

Aspect	Current assumption
Deployment	Static, single-floor, no walls; one stationary candidate and stationary background stations.
Radios	Homogeneous IEEE 802.11n devices at 5 GHz with 20 MHz channels and Minstrel-HT rate control [5].
Propagation	Yans Wi-Fi channel; log-distance loss (exponent 3.0, 46.6777 dB reference loss at 1 m) followed by Nakagami fading.
Traffic	Uplink UDP; constant background offered rates and a saturated candidate. Wired backhaul is not a bottleneck.
Observation	Passive and client-side; no probe request, association, or network-provided load report during feature collection.
Target	Mean candidate uplink throughput from successful association until the simulation stops; failed association is assigned zero.

The environment is static and open, so it excludes mobility, walls, hardware diversity, and application demand that varies over time. The label is uplink UDP throughput, so it says little about TCP goodput, downlink performance, or latency.

The choice set contains only APs discovered during the projected scan. In passive mode, discovery requires at least one decoded beacon, and observations with fewer than two discovered APs are excluded because no selection decision exists.

III. SIMULATOR DESIGN

The simulator is built on ns-3.45 and its Wi-Fi models [6]. APs sit 30 m apart in a fixed layout: two form a pair, three an equilateral triangle, and four, six, and eight fill 2×2 , 3×2 , and 4×2 grids. Channels 36, 40, 44, and 48 are assigned in

AP-index order and reused once a deployment exceeds four APs. Because the layout is fixed as well, co-channel separation takes a single value per deployment size — 30 m at eight APs, 42 m at six, and no reuse at or below four — which ties channel reuse to AP count. Section VI returns to this.

Each scenario draws 2, 3, 4, 6, or 8 APs and 2, 3, or 4 background stations per AP. A quarter of scenarios spread those stations uniformly. The rest place one or more hotspot APs, with fewer hotspots more likely and the count capped at $\lceil N_{AP}/3 \rceil$. Around those hotspots 70% of background stations fall in 10 m discs, and the remainder cover the footprint. Mixing uniform and deliberately clustered regimes follows common practice in WLAN studies [3], [4], and matters here because under uniform load every AP is about equally busy and the choice collapses back to signal strength. Each station associates with its nearest AP, and every background station in a scenario offers the same load, drawn uniformly at random from 1.0, 2.5, 4.0, and 6.0 Mbit/s. Candidate positions are drawn to keep the decision non-trivial: 70% near a boundary between neighboring APs, 30% at 4–16 m from a single AP. These are sampling policies rather than radio parameters, and they are recorded with the dataset because they set its difficulty.

Background traffic starts at $t = 1$ s. Passive recording runs until $t = 5.5$ s, covering 4.5 s of loaded traffic; the candidate associates at $t = 6$ s and the run ends at $t = 18$ s. The 0.5 s gap between recording and association is a guard: ns-3 performs target-dependent polling just before a join, and without the gap that polling would land inside the observation window. After joining, the candidate offers 100 Mbit/s in 1250-byte packets, which keeps it backlogged so that measured throughput reflects the service the network gives it rather than the limits of its own traffic generator.

A. Matched Counterfactual Replays

For each sampled topology and radio seed, the experiment runs once per target AP. AP and station positions, background traffic, the candidate’s position, and every pre-association device stay fixed; only the AP selected at $t = 6$ s changes. A `topologySeed` controls background-station placement and an independent `rngSeed` controls fading, contention, and rate adaptation, with five radio seeds generated per topology. Separating the two streams is what allows stochastic radio behavior to be repeated without the physical deployment shifting underneath it.

B. Validation

Validation runs before any model is fitted, so that a good accuracy number cannot mask a defective data-generating process.

Three properties carry the experimental design, and each is checked automatically. Repeating a seed reproduces throughput exactly, so any run can be regenerated. Changing only the radio seed varies the outcome while every physical position stays fixed. And the recorded frame and channel-busy traces are byte-identical across all target-AP replays of a matched scenario —

the check that enforces the constant O required in Section I, and the most important of the three.

These establish internal consistency, not that the simulator behaves like a real network.

IV. OBSERVATION AND DATASET CONSTRUCTION

A. Recording and Projection

The candidate carries one receive-only scanner PHY per occupied channel, plus a separate PHY reserved for association. The scanner PHYs never transmit and never associate; the association PHY is parked on unused channel 149 until the decision time, so the AP the candidate will eventually join cannot alter what is heard, or consume random draws, while the observation is being collected. At $t = 5.5$ s the scanner PHYs move to the parking channel as well.

Watching every channel at once is not a claim that real clients have several radios. It is a recording device: it captures a superset of what any single radio could have heard, from which a realistic single-radio observation is cut afterwards.

Two traces are recorded. The monitor-sniffer trace holds every frame a scanner decodes, with time, frequency, BSSID, transmitter, frame category, beacon and retry flags, length, RSSI, noise, airtime, and PHY rate. The PHY-state trace gives the fraction of each 102.4 ms beacon interval in which carrier sense marked the channel busy. This clear-channel-assessment (CCA) measurement registers energy a frame-only view misses: interference, and transmissions too distant to decode. Real chipsets expose such a counter, though whether a driver surfaces it varies by platform.

The superset is then projected onto one passive radio. That radio rotates through the four channels in random order, dwelling 110 ms on each with 0.15 ms to retune, and repeats the cycle for as much of the window as it fits, so every channel is revisited many times before the decision. Frames and CCA intervals falling outside a dwell are discarded. A 110 ms dwell spans just over one 102.4 ms beacon interval, so each visit yields about one beacon from every AP on that channel. RSSI is then perturbed by uniform error within ± 3 dB and quantized to 1 dB. Both steps run after ns-3 from a recorded scan seed, so scan policy and measurement noise can be varied without repeating a simulation.

B. From Trace to Training Data

The natural representation is the trace itself: every decoded frame in order, carrying its arrival time, channel, BSSID and transmitter, class, flags, length, signal, airtime, and PHY rate. It is a time series, and we train a transformer on it directly.

The raw observation is a long sequence, and what distinguishes one AP from another does not reside at frame granularity. Both are resolved by aggregating, which is possible only for quantities that can be averaged over a set of frames. Appendix A lists those used.

Aggregation is done per AP before the data is bucketed. Frames are attributed to an AP by BSSID. Time is bucketed by channel visit: one bin per dwell, so that no bin spans a retune and aggregates traffic from two channels into a figure

representing neither. A 5.5 s window against a 110 ms dwell gives roughly 50 bins. Each bin carries two sets of quantities: one aggregated over that AP’s own frames, the other over all traffic the radio heard during the visit. The second is shared by every AP in the choice set, since one radio hears one channel at a time, and a flag records whether the channel being visited was that AP’s own. A transformer attends over bins exactly as it did over frames.

Widening the bucket to the whole window removes time altogether and collapses everything into a single row of tabular data per AP. AP-specific quantities are aggregated over that AP’s frames alone; channel-specific quantities over the channel it occupies. Gradient boosting, random forests, and ridge regression are trained on this representation.

TABLE II
WHAT EACH MODEL RECEIVES FOR ONE CHOICE SET OF N OPTIONS.

Form	Per option	Consumed by
Frames	the decoded trace, one token per frame	Transformer
Binned	~ 50 intervals $\times$ 29 measurements	Transformer
Vector	35 measurements	GBM, RF, ridge

Squashing the time axis in this way carries no comparison between options, since it aggregates over one AP alone: the row records how busy an AP is, not whether it is busier than its neighbors. A set of relative features is therefore engineered and appended, restoring for the tabular models the comparison that per-AP aggregation removed. A sequence or set model needs no such repair, because its set stage attends across options directly.

The five radio-seed realizations of a topology are near-duplicates, so train/test splits are cut by topology: letting two of them fall on opposite sides would inflate every result reported below.

V. MODELS, PROTOCOL, AND RESULTS

A. What Is Compared

The two observations differ only in channel coverage. The *rotating* observation is the single-radio projection of Section IV; the *continuous* observation keeps the parallel-scanner superset, in which every channel is watched throughout. No single radio can do the latter, so it bounds what observation could provide rather than offering a deployable alternative. Measurement noise is identical in both.

The dataset holds 6449 AP options across 1500 choice sets drawn from 300 topologies, a median of 4 options per decision. The best option in a set yields 42.6 Mbit/s on average against 22.0 for one chosen at random, and the best and worst differ by 35.5 Mbit/s.

Decision functions are grouped by the representation they consume. Heuristics read the observables directly: strongest RSSI, least busy channel, and RSSI minus channel-busy fraction. Gradient boosting, random forests and ridge regression take the single-row vector. Transformers take the binned sequence and the frame trace.

Comparing a transformer against models on the vector confounds architecture with representation, because the two are given different information. Three further encodings separate them, all derived from the binned sequence: its average over time, a set of summary statistics per measurement, and the sequence flattened with bin order preserved. A tabular model given the flattened sequence receives exactly what the transformer receives, so a difference between them is attributable to architecture alone.

Each learned model is trained under two objectives. Regression predicts $\log(1 + \text{throughput})$, so its outputs are calibrated in Mbit/s. Ranking learns a distribution over the options in a choice set and optimizes the decision directly, at the cost of scores that carry no scale. Both are reported because they need not agree on which model wins.

Every model sees the same five splits, cut by topology and stratified by AP count. Split seeds and weight-initialization seeds are drawn independently, so the reported spread is not confounded between them. Metrics are top-1 accuracy with a 0.5 Mbit/s tie tolerance and mean regret against the oracle choice.

B. Results

Table III reports top-1 accuracy as mean and standard deviation over the five splits. Rows are grouped by the representation each model consumes; differences across columns measure what widening the observation is worth. Frame-level rows have no continuous entry, since a frame trace is what one radio received and widening it to all channels is undefined.

TABLE III
TOP-1 ACCURACY BY MODEL, OBJECTIVE, AND OBSERVATION.

Model	Obj.	Rotating	Cont.	Gain
Random choice	—	0.282	0.282	+0.000
Least busy channel	—	0.442	0.446	+0.003
Strongest RSSI	—	0.601	0.606	+0.005
RSSI minus busy	—	0.815	0.812	−0.003
Ridge, flattened sequence	reg.	0.768	0.887	+0.119
MLP, flattened sequence	rank	[−]	[−]	[−]
MLP, summary statistics	rank	[−]	[−]	[−]
GBM, summary statistics	reg.	[−]	[−]	[−]
GBM, flattened sequence	reg.	[−]	[−]	[−]
Transformer, binned	reg.	0.919	0.920	+0.001
Transformer, binned	rank	0.931	0.929	−0.002
Transformer, frame-level	reg.	[−]	—	—
Transformer, frame-level	rank	[−]	—	—

Two ablations accompany the table. Supplying every tabular model with the mean of the other options and its own deviation from that mean hands it the cross-option comparison the set stage performs, isolating how much of any transformer margin is attributable to it. Shuffling the bins within each option, leaving bin position intact, removes temporal order while preserving everything else.

VI. THE VALUE OF SEQUENCE MODELS

A sequence model earns its parameters only if the ordering of the observation carries information its summary does not.

Two conditions are necessary. The medium must vary within the observation window, and the label must depend on what that variation implies about the period after the decision, rather than on the mean the window already reports.

Neither holds here. Within an option's own channel, after background traffic begins, the ratio of within-series to between-series standard deviation is 0.09 for frame rate, 0.09 for channel-busy fraction, and 0.17 for the option's own activity. Variation between APs exceeds variation within an AP over time by an order of magnitude. For a stationary process observed under measurement noise the sample mean is a sufficient statistic, so ordering is discardable by construction rather than by inference. The shuffle ablation tests this directly, permuting bins within an option while leaving bin position intact.

This conclusion does not rest on the intraclass correlation. That statistic is 0.988, meaning the label is nearly determined once topology and target AP are fixed, but it describes the reproducibility of the label across radio seeds and implies nothing about which representation recovers it. A situation may be fully determined by where the client stands and still be legible only from how the medium evolves. Evidence about representation must come from the observation, which is why the stationarity measurement is the relevant quantity.

Table III supports the same reading from a different direction. Widening the observation from one rotating radio to four simultaneous ones raises ridge regression on the flattened sequence by 0.119 while leaving the transformer unchanged. Widening the observation therefore does not produce better decisions; it lowers the model capacity required to reach the same decision. What the transformer supplies under rotating observation is tolerance of a gapped, unevenly sampled trace, not the ability to read a trajectory. Against a two-term heuristic that subtracts channel-busy fraction from RSSI, its margin is roughly eleven points.

The practical conclusion is that this environment cannot discriminate between architectures, because it does not contain the structure that would separate them. That is a property of the simulator, and the changes below follow from it.

A. What Simulator 2 Must Change

The environment must become non-stationary on a timescale shorter than the observation window, and visibly so in a passive trace: load that arrives and departs, stations that join and leave, interference that appears part-way through. Equally important, the label must stop being an average over a long horizon. A twelve-second mean of a process that is already stationary is predicted best by the window mean; a near-horizon label, measured over the second or two immediately after the decision, is what makes a trend worth extrapolating.

Four smaller changes follow from the present work. Channel assignment by AP index over a fixed layout leaves co-channel separation with no variance and ties reuse to deployment size; sampling the assignment would decouple them. AP spacing of 30 m pairs an obstructed-indoor path-loss exponent with an open-field geometry. The BSS Load element, which a passive client reads from a single beacon, is absent and is the strongest

baseline this comparison lacks. And the restriction to complete silence, rather than to the pre-association period, could be relaxed to admit probe requests.

B. Which Question Survives Contact With Reality

The two objectives are not equally testable outside a simulator, and the difference favors the one this work treats as secondary.

Ranking asks which AP is best. Establishing ground truth for that answer requires joining each candidate under identical conditions, which is exactly what the matched replay of Section I provides and what no field measurement can. A real deployment cannot be rewound: by the time a client has joined the second AP, a microwave has started, a neighbor has begun a download, and the situation the first measurement described no longer exists. A ranking model's output is therefore unfalsifiable in the field, however well it scores here.

Regression asks what throughput a given AP would deliver. Validating that needs one association and one measurement, both of which a real client already performs. The prediction is checkable against the outcome it names, and errors accumulate into a calibration curve over ordinary use.

This work reports both objectives because the simulator supports both. A successor aimed at deployment should treat throughput prediction as the primary target, and derive the choice from it, rather than optimizing the choice directly and inheriting a claim that cannot be checked.

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APPENDIX A HAND-BUILT FEATURES

The 35 columns available to the tabular models, all derived from frames the candidate decoded before transmitting or associating. Prefixes are dropped for space. Every rate and fraction is normalized by time actually spent listening to the relevant channel, not by the observation window.

TABLE IV

THE FEAT_ COLUMNS, AP-LEVEL AND CHANNEL-LEVEL FEATURES DESCRIBE THE OPTION; RELATIVE FEATURES DESCRIBE IT AGAINST ITS RIVALS, AND EXIST TO RESTORE COMPARISONS THAT PER-AP FILTERING REMOVED.

Column	Meaning
AP-level	
ap_rssi_mean	Mean beacon RSSI (dBm)
ap_rssi_std	Spread of beacon RSSI
ap_rssi_max	Strongest beacon seen
ap_rssi_min	Weakest beacon seen
ap_rssi_last	Most recent beacon RSSI
ap_beacons	Beacons decoded
ap_beacon_gap_mean	Mean interval between decoded beacons; set by scan revisit, not beacon period
ap_beacon_gap_std	Spread of that interval
ap_frames	Frames attributed to this BSS
ap_frames_per_s	The same, per second listening
ap_bytes_per_s	Bytes per second attributed to this BSS
ap_airtime_frac	Fraction of listening time this BSS occupied
ap_n_clients	Distinct transmitters other than the AP
ap_data_frames	Data frames from this BSS
ap_mean_data_rate	Mean PHY rate of its data frames
ap_retry_frac	Fraction of its frames marked retry
Channel-level	
chan_busy_frac	Channel occupancy from decoded frame airtime
chan_cca_busy_frac	Channel occupancy from PHY carrier sense; counts energy never decoded
chan_frames_per_s	All frames per second on this channel
chan_bytes_per_s	All bytes per second
chan_data_frac	Share of frames that are data
chan_mgmt_frac	Share that are management
chan_ctrl_frac	Share that are control
chan_retry_frac	Share marked retry
chan_mean_data_rate	Mean PHY rate of data frames
chan_n_bssids	Distinct BSSIDs heard
chan_n_tas	Distinct transmitters heard
chan_mean_rssi	Mean RSSI over all frames
Relative	
rel_n_options	Options in this choice set
rel_rssi_rank	Options strictly stronger than this one
rel_is_strongest	Whether no option is stronger
rel_rssi_margin_best_other	RSSI minus the best alternative's
rel_rssi_minus_mean	RSSI minus the option-set mean
rel_airtime_share	Share of observed airtime; denominator spans every BSS heard, so it does not sum to one
rel_clients_share	Share of observed transmitters